

XAUUSD MTF BB Pullback v2.5

System Specification & Overview

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1. SYSTEM OVERVIEW

Instrument	XAU/USD — Gold vs US Dollar
System Type	MTF BB Pullback — Semi-auto Signal Generator
Broker	Exness MT5
Account	\$100 Risk per trade: 1% (\$1.00)
Tech Stack	GitHub Actions + TwelveData API + Claude Sonnet 4.6 + Telegram + Notion

2. TIMEFRAME STRUCTURE

Trend Timeframe — 1H

EMA21 vs EMA50 → defines BULLISH / BEARISH direction
ADX(10) ≥ 22.0 → confirms trend strength
 $|EMA21-EMA50| / EMA50 > 0.1\%$ → rejects flat / sideways market

Entry Timeframe — 5m

BB(20, 2.0) → identifies price overextension at band extremes
RSI(14) → confirms momentum direction is turning
ATR(14) → measures volatility; used for SL, TP, lot size, and activity filter

3. INDICATOR PARAMETERS

Bollinger Bands	Period = 20 StdDev = 2.0
RSI	Period = 14
ATR	Period = 14
EMA	Fast = 21 Slow = 50
ADX	Period = 10
Regime	Trend if ADX ≥ 25 Range if ADX < 25

4. SIGNAL CONDITIONS

BUY Signal — all 8 conditions must be true

#	Condition	Note
1	1H: EMA21 > EMA50	Bullish trend
2	1H: ADX ≥ 22.0	Strong trend
3	1H: $ EMA21-EMA50 / EMA50 > 0.1\%$	Not flat
4	5m: prev_low < BB_lower	Breached lower band
5	5m: current_close > BB_lower	Closed back inside
6	5m: prev_RSI < 48 AND RSI > prev_RSI	Momentum turning up

7	5m: $\text{ATR} \geq \text{ATR_avg_20bar} \times 1.2$	Market active
8	$\text{SL distance} \leq 3.0 \times \text{ATR}$	SL not too wide

SELL Signal — all 8 conditions must be true

#	Condition	Note
1	1H: $\text{EMA21} < \text{EMA50}$	Bearish trend
2	1H: $\text{ADX} \geq 22.0$	Strong trend
3	1H: $ \text{EMA21} - \text{EMA50} / \text{EMA50} > 0.1\%$	Not flat
4	5m: $\text{prev_high} > \text{BB_upper}$	Breached upper band
5	5m: $\text{current_close} < \text{BB_upper}$	Closed back inside
6	5m: $\text{prev_RSI} > 52$ AND $\text{RSI} < \text{prev_RSI}$	Momentum turning down
7	5m: $\text{ATR} \geq \text{ATR_avg_20bar} \times 1.2$	Market active
8	$\text{SL distance} \leq 3.0 \times \text{ATR}$	SL not too wide

5. RISK MANAGEMENT

Risk/Trade	1% of account balance (\$1.00 on \$100 account)
Lot formula	$\text{floor}(\text{balance} \times 0.01 / (\text{SL_dist} \times 100) / 0.01) \times 0.01 \mid \text{min } 0.01$
BUY SL	$\text{prev_low} - 0.3 \times \text{ATR}$
SELL SL	$\text{prev_high} + 0.3 \times \text{ATR}$
Max SL	$\text{SL distance} \leq 3.0 \times \text{ATR}$ (signal rejected if wider)
Take Profit	$\text{Entry} \pm 2.5 \times \text{SL distance} \rightarrow \text{Risk:Reward} = 1 : 2.5$
Break-even	$\text{Entry} \pm 1.0 \times \text{ATR}$
Signal expiry	10 minutes time-based OR $0.5 \times \text{ATR}$ price-based

6. AUTO-SKIP FILTERS

Weekend	Saturday all day Sunday all day Friday 23:00+ UTC
Rollover	21:59 – 22:10 UTC daily
Session guard	Active only 06:00 – 18:00 UTC (London + New York sessions)
Loss streak	Pause if ≥ 3 consecutive losses TODAY (UTC) $\rightarrow$ auto-resets next day
Weak trend	Skip if $\text{ADX} < 22.0$
Low volatility	Skip if $\text{ATR} < \text{ATR_avg} \times 1.2$
Flat market	Skip if $\text{EMA spread} < 0.1\%$

7. SYSTEM FLOW

GitHub Actions (cron: Mon–Fri 06:00 UTC & 12:00 UTC)
fetch_ohlcv.py $\rightarrow$ 1H 55 bars + 5m 55 bars via TwelveData API
fetch_calendar.py $\rightarrow$ ForexFactory USD High-impact events ± 2 hours

signal_filter.py → Rule-based check (8 conditions per direction)

FAIL → log NO_SIGNAL → sleep 300s → next iteration

PASS → context_builder.py → format structured context

analyst.py → Claude Sonnet 4.6 API → parse JSON verdict

telegram.py → send formatted signal alert

notion.py → create page in Notion database

8. CLAUDE AI ROLE

Claude acts as **Senior Quantitative Strategist for XAUUSD**. Receives full context: indicators, 1H×20 bars + 5m×50 bars price action, economic calendar (±2 hrs), and recent trade history.

Model	claude-sonnet-4-6 max_tokens: 1000
verdict	GO / NO-GO / CAUTION
confidence	HIGH / MEDIUM / LOW
reasoning	2–3 sentence price action analysis
key_levels	Important price levels to watch
risk_flags	Identified risks (news, structure, etc.)
suggested_adjustment	none / reduce_lot / widen_sl / tighten_sl
news_risk	none / low / high
Fallback	Invalid JSON or API error → treated as NO-GO automatically

9. GITHUB ACTIONS SCHEDULE

Trigger	Cron Mon–Fri + manual workflow_dispatch
Job 1	06:00 UTC (13:00 Thailand) → London session open
Job 2	12:00 UTC (19:00 Thailand) → NY session open
Runtime	355 minutes per job (~6 hours continuous)
Loop interval	Every 300 seconds (5 minutes)
Coverage	London 06–12 UTC + NY 12–18 UTC

10. NOTION DATABASE SCHEMA

Database ID: be871b32ef3549068c9d3dea76d3fbca

Auto-filled by bot:

Field	Type	Value
Signal	Title	'{DIR} @ {entry} {timestamp}'
Timestamp	Date	UTC datetime
Direction	Select	BUY / SELL
Entry / SL / TP	Number	Price values

SL Dist / Lot	Number	Calculated
Session	Select	London / NY / NY_Late / Asia
Regime	Select	trend / range
ADX / RSI / ATR	Number	Indicator snapshot
Claude Verdict	Select	GO / NO-GO / CAUTION
Claude Confidence	Select	HIGH / MEDIUM / LOW
Claude Reasoning	Text	AI analysis
Claude Risk Flags	Text	Comma-separated
Claude Adjustment	Select	none / reduce_lot / widen_sl / tighten_sl
News Risk	Select	none / low / high
Action	Select	CONFIRM (default auto)

Fill manually after each trade:

Field	Type	Value
Action	Select	CONFIRM / SKIP / EXPIRED
Actual Entry	Number	Real fill price
Exit Price	Number	Closing price
PnL	Number	Profit/Loss in USD
Hold Time Min	Number	Minutes position was held

11. FUTURE OPTIMIZATION NOTES

ADX threshold	22.0 → raise to 25+ for stronger trend-only entries
RSI thresholds	48/52 → tighten for more selective entries
ATR multiplier	1.2× → increase to filter low-volatility chop
TP ratio	2.5R → backtest 2.0R vs 3.0R for win-rate vs expectancy
BB parameters	20/2.0 → try 14/1.5 for higher signal frequency
Session filter	Narrow to London open 06–09 UTC for cleanest setups
MT5 integration	Windows VPS + MetaTrader5 Python lib for auto-execution
Loss streak logic	Consider equity-based pause instead of count-based
Multi-instrument	Extend to EURUSD / GBPUSD with same framework